

RESEARCH PROJECT REPORT

under

INSPIRE – Scholarship for Higher Education (SHE)

IDENTIFICATION AND EXTRACTION OF STEP-FLING PULSE

Submitted by

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Project Duration: 8 weeks

Date of Start: 02/06/2026

Date of Completion: 30/07/2026

. PROJECT COMPLETION CERTIFICATE

This report is submitted in connection with the INSPIRE–SHE research project entitled “IDENTIFICATION AND EXTRACTION OF STEP-FLING PULSE”, carried out by SAKHYA SHARMA under the supervision of Dr. Kshitij Kumar Yadav, Assistant Professor, Department of Civil Engineering, IIT (BHU), Varanasi. The project was undertaken for a period of 8 weeks, from 02/06/2026 to 30/07/2026.

The report follows the research-project structure prescribed in the supplied INSPIRE project-report format. The scientific discussion is based on the project information recorded in the supplied Project Completion Certificate and on the stated use of PEER ground-motion records.

3. Acknowledgement

I sincerely express my gratitude to my research supervisor, Dr. Kshitij Kumar Yadav, Assistant Professor, Department of Civil Engineering, IIT (BHU), Varanasi, for his guidance, technical suggestions and encouragement throughout this research project.

I also acknowledge the support provided by the Department of Civil Engineering, IIT (BHU), Varanasi, and the Pacific Earthquake Engineering Research (PEER) Ground Motion Database for providing access to ground-motion records used for the study.

Finally, I thank my institute, teachers, colleagues and everyone who supported me during the completion of this project.

4. Title of Research Project

IDENTIFICATION AND EXTRACTION OF STEP-FLING PULSE.

5. Aim/Objectives

The primary aim of the project is to identify and extract fling-step pulse characteristics from near-fault earthquake ground-motion records and to understand their significance for earthquake engineering.

Objectives: To understand the physical origin and engineering significance of fling-step ground motion; obtain and organize PEER acceleration records; examine H1, H2 and vertical components from 101 RSN stations; derive velocity and displacement histories; identify one-sided long-period velocity-pulse characteristics; distinguish permanent/static displacement from transient shaking; and develop a reproducible data-processing workflow.

6. Introduction of Research Topic

Near-fault earthquake ground motions can contain distinctive long-period pulses that may impose large displacement demands on structures. One important mechanism is the fling-step effect, associated with permanent tectonic offset across a rupturing fault.

A fling-step record is characterized, in an idealized form, by a one-sided pulse in ground velocity and a non-zero permanent displacement after the main shaking. A practical difficulty is that permanent displacement is sensitive to baseline errors and low-frequency processing.

In this project, earthquake time histories were examined from the PEER database. The study considered 101 RSN stations, with H1, H2 and Vertical components for each station. Acceleration histories were converted to velocity and displacement histories to investigate the signatures required for fling-step identification.

7. Theoretical Framework of Topic

The theoretical framework is based on the relationship among ground acceleration, velocity and displacement.

$$v(t) = v(t_0) + \int_{t_0}^t a(\tau) d\tau$$

$$u(t) = u(t_0) + \int_{t_0}^t v(\tau) d\tau$$

The permanent displacement associated with a fling-step motion can be represented by the difference between the stable displacement level after the event and the initial reference level. In practical records, this quantity must be interpreted only after appropriate baseline and filtering checks.

A conceptual decomposition is $a(t) = a_p(t) + a_r(t)$, where $a_p(t)$ is the pulse/fling-related contribution and $a_r(t)$ is the remaining transient ground motion. The velocity history is examined for a dominant one-sided pulse and the displacement history for a stable non-zero offset.

8. Profile of Organization/Research Lab

The research work was carried out under the Department of Civil Engineering, Indian Institute of Technology (BHU), Varanasi. The department provides an academic and research environment covering structural engineering, earthquake engineering, geotechnical engineering, transportation engineering, water resources and related civil-engineering disciplines.

The relevant research area for this project was structural and earthquake engineering, with emphasis on interpretation and processing of strong ground-motion records. The project was supervised by Dr. Kshitij Kumar Yadav.

The PEER Ground Motion Database was used as the principal external data source for searching, selecting and downloading earthquake ground-motion data for engineering research.

9. Methodology Followed

1. Select earthquake ground-motion records from the PEER database.
2. Organize records corresponding to 101 RSN stations.
3. Separate and examine H1, H2 and Vertical components.
4. Read acceleration time histories and construct the corresponding time vector using the record sampling interval.
5. Plot acceleration versus time and inspect amplitude, duration and major phases of shaking.
6. Numerically integrate acceleration to obtain velocity and velocity to obtain displacement.
7. Check baseline errors, low-frequency drift and unrealistic displacement trends.
8. Identify candidate one-sided velocity pulses and examine the corresponding residual/permanent displacement.
9. Document relevant pulse characteristics for engineering interpretation.
10. Compare H1, H2 and vertical components to assess directional dependence.

10. Analysis and Interpretation of Research Project

The analysis focused on time-domain characteristics. Acceleration histories provide information about high-frequency shaking and peak acceleration, velocity histories are more useful for recognizing long-period pulse behavior, and displacement histories provide direct evidence for a permanent ground offset but are highly sensitive to baseline errors.

For each record, the acceleration time series was inspected for data quality and duration. The integrated velocity history was examined for a dominant one-sided pulse. A candidate fling-step motion should show a sustained velocity excursion of one principal sign rather than only a symmetric oscillatory pulse.

The displacement history was then examined. A fling-step signature is supported when displacement evolves toward a new, approximately stable level after the principal shaking. This interpretation must be combined with baseline-correction and filtering checks because integration can amplify small low-frequency errors.

The 101-station dataset and its three components allow the horizontal directions to be assessed independently while the vertical motion is treated separately. This comparison is important because fling-step effects are related to fault geometry and can be much more pronounced in one horizontal direction.

The generated time-history plots provide a direct connection between peak acceleration, velocity-pulse amplitude, pulse duration and final displacement. These observations are useful for selecting records for later nonlinear

structural analysis.

11. Conclusion & Suggestion of Research Project

The project established a systematic workflow for identification and extraction of fling-step pulse characteristics from earthquake ground-motion records. It demonstrates that acceleration, velocity and displacement histories should be interpreted together rather than relying on acceleration peaks alone.

The key diagnostic features are a one-sided, long-period velocity pulse and a corresponding non-zero permanent displacement after the main shaking. Baseline errors and inappropriate low-frequency processing can create or remove apparent permanent displacement; therefore processing quality is essential.

Future work can use the extracted pulse parameters to classify records quantitatively, compare fling-step and forward-directivity motions, investigate pulse-period and amplitude relationships, and evaluate the influence of fling-step motions on structural response. Automated signal processing and statistical classification can also be developed.

12. Result(s) Achieved

1. Developed a structured understanding of the physical mechanism and engineering significance of fling-step ground motion.
2. Organized and examined ground-motion records from 101 RSN stations.
3. Performed component-wise analysis of H1, H2 and Vertical records.
4. Generated/interpreted acceleration, velocity and displacement time histories.
5. Identified the principal time-domain indicators used to recognize candidate fling-step motions.
6. Studied the influence of baseline errors and low-frequency processing on residual displacement.
7. Developed a repeatable data-analysis workflow suitable for extension to a larger dataset.
8. Improved practical understanding of PEER earthquake records and their application in earthquake-engineering studies.

13. References

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4. USGS (2014). Adding fling effects to processed ground-motion time histories.
5. Daei, A., Poursha, M. and Zarrin, M. (2023). Sensitivity of structural responses to the processing of near-fault ground motion records containing fling-step.
6. Chang, Z., Li, W., Goda, K. and Yan, Z. (2025). Automated Recovery of Permanent Displacement in Near-Fault Ground Motions with Fling-Step Effects. *Earthquake Engineering & Structural Dynamics*, 54(14), 3685–3704.

14. Declaration by the Scholar

I, SAKHYA SHARMA, hereby declare that the details and scientific description presented in this research project report are true to the best of my knowledge and have been prepared on the basis of the project work carried out under the supervision of Dr. Kshitij Kumar Yadav.

I understand that any discrepancy in the information submitted in connection with the INSPIRE research project shall be my responsibility.

Date: 07/07/2026

Place: IIT, BHU, Department of Civil Engineering

Signature of Scholar: _____

SAKHYA SHARMA

Supervisor: _____

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